Filippo Santoliquido

email: filippo.santoliquido@gssi.it

website: https://filippo-santoliquido.github.io/

Academic positions

- **Postdoctoral researcher**, *Gran Sasso Science Institute (GSSI)*, L'Aquila, Italy. May 2025 - now
- **Postdoctoral researcher**, *Gran Sasso Science Institute (GSSI)*, L'Aquila, Italy. Sep. 2023 - May 2025
  - I am a postdoctoral researcher at GSSI since September 2023. I am working within the group led by Prof. Marica Branchesi. My work focuses on developing advanced tools for population inference and data analysis for gravitational-wave interferometers, with a special focus on the Einstein Telescope. In May 2025, I began a new type of postdoctoral contract funded by the European Union's *NextGenerationEU* program, administered through the Italian Ministry of University and Research. This was part of a call for recruiting international postdoctoral researchers, established by Decree No. 47 of February 20, 2025 (D.R. n. 76/2025), with CUP (unique project code) D13C25000700001.
- **Postdoctoral researcher**, *University of Padova*, Padova, Italy. Jan. 2023 - Sep. 2023
  - I conducted postdoctoral research at the University of Padova within Prof. Michela Mapelli's group. My work focused on simulating the population properties of compact object mergers—an essential step for interpreting gravitational-wave observations with the LIGO–Virgo–KAGRA detectors and for developing the scientific case of the Einstein Telescope.

Education

- **PhD student in Astrophysics**, *University of Padova*, Oct. 2019 - 3 Apr. 2023 (defence date).
  - During my PhD, I spent six months visiting Dr. Stanilav Babak at the *Astroparticule et Cosmologie (APC) laboratory*, Paris, France. Feb. - Jul. 2022.

- **Master Student in Astronomy**, *University of Padova*, 110/110, Oct. 2017 - 26 Sep. 2019 (defence date)
- **Bachelor Student in Physics**, *University of Trento*, 104/110, Sep. 2014 - 25 Sep. 2017 (defence date)

Awards and Grants

- **Seal of Excellence**, The project proposal *Neural-GW: "Neural Parameter Estimation for Next-Generation Gravitational-Wave Detectors"*, submitted under the 2025 MSCA Postdoctoral Fellowships, was evaluated by an international panel of independent experts and recognized as a high-quality project in a highly competitive process. Unfortunately, it could not be funded due to budgetary constraints. 17 Feb. 2026.
- **Tacchini Prize**, 2k euros, Vincitori dei premi SAIt 2023 | Società Astronomica Italiana. 25 May 2023
- **Erasmus+ Trainee Program**, *APC laboratory*, Paris, France, 2.1k euros. Feb. 2022 - Jul. 2022
- **2018 Merit award**, *University of Trento*, 4k euros. 10 Jun. 2019
- **Erasmus+ Program***, *University of Coimbra*, 2k euros. Feb. 2017 - Jul. 2017

Affiliations and Collaborations

- I am a member of the Observational Science Board (OSB) of the Einstein Telescope collaboration. Jun. 2022 - now. I regularly contribute in the Division 3 (“Population Studies”) and Division 10 (“Data Analysis”). A summary of my contributions:
    - I am one of the four chairs of Division 10.
    - I coordinate, together with Fabio Antonini (*Cardiff University*), the **high-redshift task force** inside the Division 3. This task force aims to bring together researchers working on population modeling and data analysis methods, with particular focus on science cases that are unique to the Einstein Telescope, such as high-redshift systems.
    - I have contributed to two key OSB documents: the Blue Book, where I worked on the section dedicated to host galaxies and served as a referee for the chapter on data analysis.
    - I also contributed to Branchesi et al., 2023, providing the source populations used to compare different proposed designs of the Einstein Telescope.

- I am an active member of the LIGO–Virgo–KAGRA collaboration Sep. 2023 – present. I regularly contribute to the Compact Binary Coalescence (CBC) division, in particular to the Rates and Population (R&P) group and the Parameter Estimation (PE) group. I contribute to the analysis of data from the fourth observing run (O4), focusing on population studies to understand the properties of compact objects, such as their mass distribution, and I serve as a referee for collaboration papers. I served as reviewer of the figures of GWTC-5 paper
- I am a community member of the LISA consortium Apr. 2025 – present
- I am a member of the GRAWITA (GRAvitational Wave INAF TeAm) collaboration Dec. 2025 – present
- I am affiliated with *INFN (Istituto Nazionale di Fisica Nucleare)*:
    - *Laboratori Nazionali del Gran Sasso (LNGS)*. Sep. 2023 - now
    - *Sezione di Padova*. Oct. 2019 - Aug. 2023
- I have been a member of TEONGRAV Oct. 2019 – May 2025

Teaching and Mentoring

Teaching

- May 2026, **Foundations of Gravitational-Wave and Multimessenger Astrophysics** (GC-9). I taught a course on gravitational-wave data analysis to the PhD students of the 41st cycle in Astroparticle Physics at GSSI. The course covered the main approaches, from the Fisher information matrix approximation, to the full Bayesian inference problem, to deep-learning-based methods for fast inference. Each lecture was followed by a hands-on session on using the codes `GWFish`, `Bilby`, and `Dingo`. Teaching material available here upon request (6 hours).

Invited lecturer

- Mar. 23, 2026, **Hands-on session on Bayesian inference**. I gave a lecture on using `Bilby` for standard inference problems, with a worked example on simplified gravitational-wave data analysis. The lecture was part of the course "Probabilistic Inference and Forecasting in the Sciences", held by Dr. Stefano Piacentini for the first-year PhD students in Physics at Sapienza University of Rome. Material available here.
- Lecturer of the course “Gravitational Wave Astrophysics” at the school “Cosmological History: from Gravitational Waves to Exoplanets”, *ICTP-SAIFR/IFT-UNESP*, São Paulo, Brazil, 4 h.

Mentoring

- Co-supervisor of **PhD thesis**:
    - Ludovico Alessio De Santis, *Gran Sasso Science Institute*, expected PhD defence Dec. 2026
    - Mirko Pitzalis, *University of Cagliari*, expected PhD defence Dec. 2026
- Co-supervisor of **master thesis**:
    - Lorenzo Merli, *University of Padova*, Mar. 2022
    - Roberta Rufolo, *University of Padova*, Dec. 2021

Teaching assistant

- Physics Lab assistant, *University of Padova*, Padova, Italy, 20 h. Nov. 2021
- Laboratory of Computational Astrophysics, *University of Padova*, Padova, Italy, 12 h. Jun. 2021
- Physics Lab assistant, *University of Padova*, Padova, Italy, 20 h. Nov. 2020

Referee

- Monthly Notices of the Royal Astronomical Society (MNRAS).
- The Astrophysical Journal (ApJ).
- Physical Review D (PRD).
- Physical Review Letters (PRL).
- Classical and Quantum Gravity (CQG)

Public Outreach

1. **Periferie creative**, a project organized by the Orbeat collective and GSSI, aimed at regenerating local urban spaces such as neighborhood parks into inclusive, participatory places. As part of the event, we brought a solar telescope and a Michelson interferometer to introduce the basics of gravitational-wave detection, L'Aquila, Italy. May 30, 2026
2. Lecture on gravitational-wave astrophysics as part of a teacher training program (Programma INFN per Docenti, PID), *LNGS*, L'Aquila, Italy. 24-29 Nov. 2025

3. Space Explorer – an outreach activity dedicated to the youngest astro-curious students at “Gianni Di Genova” Elementary School, L’Aquila, Italy. 12 May 2025

4. Night of Researchers, volunteer at European researcher night - SHARPER, *GSSI and LNGS*, L’Aquila, Italy. 27 Sep. 2024

5. Public outreach lecture and stargazing activities during the public event "Tutte le direzioni della luce", Sant'Antioco, Italy. 19 Sep. 2024

6. Outreach lecture on gravitational-wave astrophysics and future prospects at the “A. Volta” High School. 27 Mar. 2024

7. Career orientation lecture for third-year students at the Einstein High School in Teramo, *GSSI*, L'Aquila, Italy 23 Jan. 2024

8. Night of Researchers, volunteer at European researcher night - SHARPER, *GSSI and LNGS*, L’Aquila, Italy. 29 Sep. 2023

9. Night of Researchers, *University of Padova*, Virtual meeting. 27 Nov. 2020

10. Guide for the Museum of History of Physics, *University of Padova*. Oct. 2018 - Mar. 2020

Technical Skills

	Programming and scripting
Advanced	Python, Jupyter notebooks, Latex
Intermediate	BASH scripting, Parallel computing (CPUs and GPUs)
Basic	PyTorch, Keras, MatLab, R, C++, SQL
	Operating Systems
Advanced	Linux, iOS, Windows
	Versioning and Cloud
Advanced	Git - see public repositories
	Main developer
	cosmo$\mathcal{R}$ate
	galaxy$\mathcal{R}$ate
	ET_classifier

I also usually run on High Performance Computing (HPC) machines, such as the servers of the LIGO data grid

Schools and Programs

1. Introduction to Deep Learning, CINECA, Rome, Italy. 10-12 Apr. 2024
2. ML-INFN Hackathon: Advanced Level, *INFN-Bari*, Italy. 21-24 Nov. 2022
3. Hands-on machine learning course with Python, *OAPd*, Padova, Italy. Oct. 2022
4. First ML-INFN Hackathon, *INFN-Firenze-Pisa-CNAF*, Italy. 7-9 Jun. 2021
5. Summer School in Statistics for Astronomers XVI, *Penn State University*, USA. 1-5 Jun. 2021
6. Gravitational waves: a new messenger to explore the universe, *IHP*, Paris, France. 1 Mar.-9 Apr. 2021
7. SIGRAV school, Rome, Italy. 1-5 Feb. 2021

8. Multi Messenger Astrophysics School, Asiago, Italy. 14-23 Jan. 2020

Languages

Italian Native speaker, **English** Excellent

Invited Colloquia and Seminars

1. Invited speaker, *Marietta Blau Institute*, Vienna, Austria. 23 Jan. 2026
2. Invited speaker, *Trento Institute for Fundamental Physics and Applications (TIFPA)*, Trento, Italy. 4 Dec. 2025
3. Invited speaker, *Johns Hopkins University*, Baltimore, USA. 10 Jun. 2025
4. Invited speaker, GW-BO meeting, *University of Bologna*, Bologna, Italy. 11 Feb. 2025
5. Invited speaker, *INFN-Cagliari*, Cagliari, Italy. 29 Nov. 2024
6. Invited speaker, Department of Astronomy, *University Andrés Bello*, Santiago, Chile. 6 Aug. 2024
7. Invited speaker, *INFN-Laboratori Nazionali del Gran Sasso*, L'Aquila, Italy. 22 Mar. 2024
8. Invited speaker, Department of Physics, *University of Milano-Bicocca*, Milan, Italy. 22 Sep. 2022
9. Invited speaker, High Energy Group Meeting, *Institut d'Astrophysique de Paris (IAP)*, Paris, France. 24 Jun. 2021
10. Invited speaker, GW group meeting, *Observatoire de la Côte d'Azur*, Nice, France. 29 May - 2 Jun. 2022
11. Invited speaker, *SISSA Astrophysics Colloquium*, Trieste, Italy. 13 Apr. 2021

Conferences and Workshops

1. Contributed talk at "AI for Gravitational Wave workshop", CERN, Geneva, Switzerland. 5-8 May 2026
2. **Invited speaker** (on behalf of the LIGO-Virgo-KAGRA collaboration) at "GRaviCon 2026", Pisa, Italy. 22-24 Apr. 2026

3. **Invited speaker** and panelist at the "Einstein Telescope science workshop for early career researchers", Rome, Italy. 18-20 Feb. 2026

4. Contributed talk at "GWFreeride: Carving the AI Gradient in Gravitational-Wave Astronomy", Sexten, Italy. 26-30 Jan. 2026

5. Contributed talk at the "4th Einstein Telescope annual meeting", Opatija, Croatia. 11-14 Nov. 2025

6. Contributed talk at "FROM DEEP SPACE TO DEEP EARTH: THE EXPERIMENTAL NETWORK IN SARDINIA - A GEOPHYSICAL WORKSHOP", Orosei, Italia. 6-10 Oct. 2025

7. **Invited speaker** and panelist at the "1st Sardinian GW Science mini-workshop", Cagliari, Italia. 24-25 Sep. 2025

8. Contributed talk at "EuCAIFCon 2025", Cagliari, Italia. 16-20 Jun. 2025

9. Poster at the workshop "Scientific Machine Learning for Gravitational Wave Astronomy", The Institute for Computational and Experimental Research in Mathematics (ICERM), Providence, USA. 2-6 Jun. 2025

10. Contributed talk at the "XV Einstein Telescope Symposium", Bologna, Italia. 26-30 May 2025

11. Contributed talk and panelist at the workshop "Challenges and future perspectives in gravitational-wave astronomy: O4 and beyond", Lorentz Center, Leiden, Netherlands. 14-18 Oct. 2024

12. Contributed talk at the annual TEONGRAV meeting, Sapienza Università di Roma, Rome, Italy. 16-20 Sep. 2024

13. Contributed talk at the 41ST LIAC, Université de Liège, Liège, Belgium. 15-19 Jul. 2024

14. Participation at the workshop "Fundamental Physics in the Era of Big Data and Machine Learning", Aspen Center for Physics, Aspen, USA. 27 May - 9 Jul. 2024

15. Contributed talk at the XIV ET Symposium, Maastricht, Netherlands. 6-10 May 2024

16. Contributed talk at the "Fourth EPS Conference on Gravitation: Black Holes", Valencia, Spain. 13-15 Nov. 2023

17. Poster at GraSP23, University of Pisa, Pisa, Italy. 24-27 Oct. 2023

18. Contributed talk at IV Gravi-Gamma-Nu workshop, Gran Sasso Science Institute, L'Aquila, Italy. 4-6 Oct. 2023

19. Panelist at GWPOPNEXT, University of Milano-Bicocca, Milan, Italy. 10-14 Jul. 2023
20. Contributed talk at XIII ET symposium, Cagliari, Italy. 8-12 May 2023
21. Contributed talk at EAS2022, annual meeting of the European Astronomical Society during the symposium: "Gravitational Wave and Multi-messenger Astronomy: current results and future perspectives", Valencia, Spain. 27 Jun. - Jul. 1 2022
22. Participation at the workshop "Bayesian Deep Learning in Astrophysics", Paris, France. 20-24 Jun. 2022
23. Contributed talk at the American Physical Society April Meeting 2022, New York City, New York, USA. 9-12 Apr. 2022
24. Contributed talk at GWday, Padova, Italy. 2 Dec. 2021
25. Contributed talk at Amaldi14, Melbourne, Australia. 19-23 Jul. 2021
26. Jun. 2021 Poster at EAS2021, annual meeting of the European Astronomical Society during the symposium: "The Birth, Life, and Death of Black Holes". Leiden, Netherlands. Jun. 28-Jul. 2 2021
27. Contributed talk at "3rd Workshop on Chemical Abundances in Gaseous Nebulae", Universidade do Vale do Paraíba, São José dos Campos, Brazil. 24-28 May 2021
28. Participation at the workshop "#4 Gravitational Wave Open Data Workshop", Virtual meeting. 10-14 May 2021
29. Contributed talk at "The fourth assembly of the Groupement de Recherche Ondes Gravitationnelles", Paris, France. 30 Mar. -1 Apr. 2021
30. Poster at 55th Rencontres de Moriond 2021, La Thuile, Italy. 9-11 Mar. 2021
31. Contributed talk at annual TEONGRAV meeting, Virtual meeting. 18 Feb. 2021
32. Poster at EAS2020, annual meeting of the European Astronomical Society during the symposium: "What have we learned from the observed population of gravitational wave sources?", Leiden, Netherlands. 29 Jun.-3 Jul. 2020
33. **Invited talk** at Mock Innsbruck, Innsbruck, Austria. 10-13 Mar. 2020

List of Publications

ADS, ORCID, Google Scholar

Summary of Publications

Total publications	75
Total publications as first author	10
Refereed publications	47
Refereed publications as first author	6
Total number of citations	4186
Total number of citations as first author	390
h-index	32
h-index as first author	5

Data taken from ADS on March 2, 2026

First-author peer-reviewed publications

1. *Fast and accurate parameter estimation of high-redshift sources with the Einstein Telescope*, Santoliquido et al., PhRvD 112(10), 11/2025, 2025PhRvD.112j3015S.

2. *Classifying binary black holes from Population III stars with the Einstein Telescope: A machine-learning approach*, Santoliquido et al., A&A 690 10/2024, 2024A&A...690A.362S.

3. *Binary black hole mergers from population III stars: uncertainties from star formation and binary star properties*, Santoliquido et al., MNRAS 524(1), 09/2023, 2023MNRAS.524..307S.

4. *Modelling the host galaxies of binary compact object mergers with observational scaling relations*, Santoliquido et al., MNRAS 516(3), 11/2022, 2022MNRAS.516.3297S.

5. *The cosmic merger rate density of compact objects: impact of star formation, metallicity, initial mass function, and binary evolution*, Santoliquido et al., MNRAS 502(4), 04/2021, 2021MNRAS.502.4877S.

6. *The Cosmic Merger Rate Density Evolution of Compact Binaries Formed in Young Star Clusters and in Isolated Binaries*, Santoliquido et al., ApJ 898(2), 08/2020, 2020ApJ...898..152S.

Short-author peer-reviewed publications

1. *Crystal Eye: All sky MeV monitor with high precision real-time localization*, Aloisio et al., APh 174 01/2026, 2026APh...17403171A.

2. *Black hole—neutron star and binary neutron star mergers from Population III and II stars*, Mestichelli et al., A&A 704 12/2025, 2025A&A...704A..54M.

3. *The more accurately the metal-dependent star formation rate is modeled, the larger the predicted excess of binary black hole mergers*, Sgalletta et al., A&A 698 06/2025, 2025A&A...698A.144S.

4. *Prospects for optical detections from binary neutron star mergers with the next-generation multi-messenger observatories*, Loffredo et al., A&A 697 05/2025, 2025A&A...697A..36L.

5. *A new prescription for the spectral properties of population III stellar populations*, Lecroq et al., A&A 695 03/2025, 2025A&A...695A..17L.

6. *Validating prior-informed Fisher-matrix analyses against GWTC data*, Dupletsa et al., PhRvD 111(2), 01/2025, 2025PhRvD.111b4036D.

7. *Gravitational waves from mergers of Population III binary black holes: roles played by two evolution channels*, Liu et al., MNRAS 534(3), 11/2024, 2024MNRAS.534.1634L.

8. *Binary black hole mergers from Population III star clusters*, Mestichelli et al., A&A 690 10/2024, 2024A&A...690A.106M.

9. *Binary Black Hole Spins: Model Selection with GWTC-3*, Périgois et al., Univ 9(12), 12/2023, 2023Univ....9..507P.

10. *Massive binary black holes from Population II and III stars*, Costa et al., MNRAS 525(2), 10/2023, 2023MNRAS.525.2891C.

11. *Pre-merger alert to detect prompt emission in very-high-energy gamma-rays from binary neutron star mergers: Einstein Telescope and Cherenkov Telescope Array synergy*, Banerjee et al., A&A 678 10/2023, 2023A&A...678A.126B.

12. *Compact object mergers: exploring uncertainties from stellar and binary evolution with SEVN*, Iorio et al., MNRAS 524(1), 09/2023, 2023MNRAS.524..426I.

13. *Science with the Einstein Telescope: a comparison of different designs*, Branchesi et al., JCAP 2023(7), 07/2023, 2023JCAP...07..068B.

14. *Perspectives for multimessenger astronomy with the next generation of gravitational-wave detectors and high-energy satellites*, Ronchini et al., A&A 665 09/2022, 2022A&A...665A..97R.

15. *Prospects for multimessenger detection of binary neutron star mergers in the fourth LIGO-Virgo-KAGRA observing run*, Patricelli et al., MNRAS 513(3), 07/2022, 2022MNRAS.513.4159P.

16. *Gravitational background from dynamical binaries and detectability with 2G detectors*, Périgois et al., PhRvD 105(10), 05/2022, 2022PhRvD.105j3032P.

17. *Host galaxies and electromagnetic counterparts to binary neutron star mergers across the cosmic time: detectability of GW170817-like events*, Perna et al., MNRAS 512(2), 05/2022, 2022MNRAS.512.2654P.

18. *The cosmic evolution of binary black holes in young, globular, and nuclear star clusters: rates, masses, spins, and mixing fractions*, Mapelli et al., MNRAS 511(4), 04/2022, 2022MNRAS.511.5797M.

19. *Compact object mergers in hierarchical triples from low-mass young star clusters*, Trani et al., MNRAS 511(1), 02/2022, 2022MNRAS.511.1362T.

20. *GW190521 formation via three-body encounters in young massive star clusters*, Dall'Amico et al., MNRAS 508(2), 12/2021, 2021MNRAS.508.3045D.

21. *New insights on binary black hole formation channels after GWTC-2: young star clusters versus isolated binaries*, Bouffanais et al., MNRAS 507(4), 11/2021, 2021MNRAS.507.5224B.

22. *Dynamics of binary black holes in low-mass young star clusters*, Rastello et al., MNRAS 507(3), 11/2021, 2021MNRAS.507.3612R.

23. *Mass and Rate of Hierarchical Black Hole Mergers in Young, Globular and Nuclear Star Clusters*, Mapelli et al., Symm 13(9), 09/2021, 2021Symm...13.1678M.

24. *Constraining accretion efficiency in massive binary stars with LIGO -Virgo black holes*, Bouffanais et al., MNRAS 505(3), 08/2021, 2021MNRAS.505.3873B.

25. *Hierarchical black hole mergers in young, globular and nuclear star clusters: the effect of metallicity, spin and cluster properties*, Mapelli et al., MNRAS 505(1), 07/2021, 2021MNRAS.505..339M.

26. *Binary black holes in young star clusters: the impact of metallicity*, Di Carlo et al., MNRAS 498(1), 10/2020, 2020MNRAS.498..495D.

27. *Dynamics of black hole-neutron star binaries in young star clusters*, Rastello et al., MNRAS 497(2), 09/2020, 2020MNRAS.497.1563R.

28. *Binary black holes in the pair instability mass gap*, Di Carlo et al., MNRAS 497(1), 09/2020, 2020MNRAS.497.1043D.

29. *An astrophysically motivated ranking criterion for low-latency electromagnetic follow-up of gravitational wave events*, Artale et al., MNRAS 495(2), 06/2020, 2020MNRAS.495.1841A.

30. *Host galaxies of merging compact objects: mass, star formation rate, metallicity, and colours*, Artale et al., MNRAS 487(2), 08/2019, 2019MNRAS.487.1675A.

31. *The properties of merging black holes and neutron stars across cosmic time*, Mapelli et al., MNRAS 487(1), 07/2019, 2019MNRAS.487....2M.

Collaboration peer-reviewed publications

1. *Black Hole Spectroscopy and Tests of General Relativity with GW250114*, Abac et al., PhRvL 136(4), 01/2026, 2026PhRvL.136d1403A.

2. *GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog*, Abac et al., ApJL 995(1), 12/2025, 2025ApJ...995L..18A.

3. *All-sky search for short gravitational-wave bursts in the first part of the fourth LIGO-Virgo-KAGRA observing run*, Abac et al., PhRvD 112(10), 11/2025, 2025PhRvD.112j2005A.

4. *GW231123: A Binary Black Hole Merger with Total Mass 190—265 $M_{\odot}$*, Abac et al., ApJL 993(1), 11/2025, 2025ApJ...993L..25A.

5. *GW241011 and GW241110: Exploring Binary Formation and Fundamental Physics with Asymmetric, High-spin Black Hole Coalescences*, Abac et al., ApJL 993(1), 11/2025, 2025ApJ...993L..21A.

6. *GW250114: Testing Hawking's Area Law and the Kerr Nature of Black Holes*, Abac et al., PhRvL 135(11), 09/2025, 2025PhRvL.135k1403A.

7. *Search for Gravitational Waves Emitted from SN 2023ixf*, Abac et al., ApJ 985(2), 06/2025, 2025ApJ...985..183A.

8. *Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*, Abac et al., ApJ 983(2), 04/2025, 2025ApJ...983...99A.

9. *A Search Using GEO600 for Gravitational Waves Coincident with Fast Radio Bursts from SGR 1935+2154*, Abac et al., ApJ 977(2), 12/2024, 2024ApJ...977..255A.

10. *Observation of Gravitational Waves from the Coalescence of a 2.5—4.5 M $_\odot$ Compact Object and a Neutron Star*, Abac et al., ApJL 970(2), 08/2024, 2024ApJ...970L..34A.

Other publications

1. *Constraining Binary Neutron Star Populations using Short Gamma-Ray Burst Observations*, Ludovico De Santis et al., arXiv 02/2026, 2026arXiv260213391L.
2. *Deep Search for Joint Sources of Gravitational Waves and High-Energy Neutrinos with IceCube During the Third Observing Run of LIGO and Virgo*, The IceCube Collaboration et al., arXiv 01/2026, 2026arXiv260107595T.
3. *WINK: Advancing X and Gamma Ray Detection Technology for Space Observations*, Tambone et al., icrc.conf 12/2025, 2025icrc.confE.857T.
4. *X-ray Binaries: A Key Corollary Science with Crystal Eye*, Tarana et al., icrc.conf 12/2025, 2025icrc.confE.594T.
5. *Comparing next-generation detector configurations for high-redshift gravitational wave sources with neural posterior estimation*, Santoliquido et al., arXiv 12/2025, 2025arXiv251220699S.
6. *Constraints on gravitational waves from the 2024 Vela pulsar glitch*, The LIGO Scientific Collaboration et al., arXiv 12/2025, 2025arXiv251217990T.
7. *Uncovering the population of compact binary mergers and their formation pathways with gravitational waves through the Einstein Telescope*, Arca-Sedda et al., arXiv 12/2025, 2025arXiv251217339A.
8. *GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures*, The LIGO Scientific Collaboration et al., arXiv 12/2025, 2025arXiv251216347T.
9. *Search for planetary-mass ultra-compact binaries using data from the first part of the LIGO--Virgo--KAGRA fourth observing run*, The LIGO Scientific Collaboration et al., arXiv 11/2025, 2025arXiv251119911T.
10. *All-sky search for continuous gravitational-wave signals from unknown neutron stars in binary systems in the first part of the fourth LIGO-Virgo-KAGRA observing run*, The LIGO Scientific Collaboration et al., arXiv 11/2025, 2025arXiv251116863T.
11. *Direct multi-model dark-matter search with gravitational-wave interferometers using data from the first part of the fourth LIGO-Virgo-KAGRA observing run*, The LIGO Scientific Collaboration et al., arXiv 10/2025, 2025arXiv251027022T.

12. *Cosmological and High Energy Physics implications from gravitational-wave background searches in LIGO-Virgo-KAGRA's O1-O4a runs*, The LIGO Scientific Collaboration et al., arXiv 10/2025, 2025arXiv251026848T.

13. *Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA's Fourth Observing Run*, The LIGO Scientific Collaboration et al., arXiv 10/2025, 2025arXiv251017487T.

14. *Directed searches for gravitational waves from ultralight vector boson clouds around merger remnant and galactic black holes during the first part of the fourth LIGO-Virgo-KAGRA observing run*, The LIGO Scientific Collaboration et al., arXiv 09/2025, 2025arXiv250907352T.

15. *GWTC-4.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation*, The LIGO Scientific Collaboration et al., arXiv 09/2025, 2025arXiv250904348T.

16. *Upper Limits on the Isotropic Gravitational-Wave Background from the first part of LIGO, Virgo, and KAGRA's fourth Observing Run*, The LIGO Scientific Collaboration et al., arXiv 08/2025, 2025arXiv250820721T.

17. *GWTC-4.0: Population Properties of Merging Compact Binaries*, The LIGO Scientific Collaboration et al., arXiv 08/2025, 2025arXiv250818083T.

18. *GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*, The LIGO Scientific Collaboration et al., arXiv 08/2025, 2025arXiv250818082T.

19. *GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients*, The LIGO Scientific Collaboration et al., arXiv 08/2025, 2025arXiv250818081T.

20. *Open Data from LIGO, Virgo, and KAGRA through the First Part of the Fourth Observing Run*, The LIGO Scientific Collaboration et al., arXiv 08/2025, 2025arXiv250818079T.

21. *Comparing astrophysical models to gravitational-wave data in the observable space*, Toubiana et al., arXiv 07/2025, 2025arXiv250713249T.

22. *All-sky search for long-duration gravitational-wave transients in the first part of the fourth LIGO-Virgo-KAGRA Observing run*, The LIGO Scientific Collaboration et al., arXiv 07/2025, 2025arXiv250712282T.

23. *The Science of the Einstein Telescope*, Abac et al., arXiv 03/2025, 2025arXiv250312263A.

24. *Correction to: Binary black hole mergers from population III stars: uncertainties from star formation and binary star properties*, Santoliquido et al., MNRAS 528(1), 02/2024, 2024MNRAS.528..954S.

25. *Erratum: Prospects for multimessenger detection of binary neutron star mergers in the fourth LIGO-Virgo-KAGRA observing run*, Patricelli et al., MNRAS 514(3), 08/2022, 2022MNRAS.514.3395P.

26. *The evolution of compact object mergers and their host galaxies across cosmic time*, Santoliquido et al., eas..conf 07/2022, 2022eas..conf..294S.

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